CLASS 8 · PART TEST-1 · STUDENT PERFORMANCE REPORT

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Campus HYD/Nallakunta DD Colony
JEE Mains 08 Oct 2026
JEE Advanced 09 Oct 2026

JEE Mains

08 Oct 2026 · 74 questions

66 / 296 — **22.3%**

1090	719	41/63	289
ALL-INDIA	STATE	CAMPUS	PROGRAM

Weekly-test average **154.4 / 300** (51.5%) → this test **22.3%** ▼ **29.2**

230 marks lost: 170 on 34 wrong answers (incl. 34 negative) · 60 on 15 skipped

JEE Advanced

09 Oct 2026 · 48 questions

89 / 171 — **52.0%**

200	134	6/63	54†
ALL-INDIA	STATE	CAMPUS	PROGRAM

Weekly-test average **68.9 / 180** (38.3%) → this test **52.0%** ▲ **13.7**

82 marks lost: 71 on 17 wrong answers (incl. 8 negative) · 10 on 3 skipped · 1 on partly-correct answers

1 Subject scorecard

Your marks in each subject, against the 63 students of this program on campus and against your own weekly-test average.

SUBJECT	MARKS	SCORE VS CAMPUS	SUBJECT AIR	CAMPUS AVG / BEST	WEEKLY AVG	CHANGE	RIGHT / WRONG / SKIP
JEE Mains 66 / 296 · 22.3%							
Maths	53/96	55%	103	37.0 / 77	64.2/100 64.2%	▼ 8.9	15 / 7 / 2
Physics	10/100	10%	1286	18.3 / 60	49.3/100 49.3%	▼ 39.3	6 / 14 / 5
Chemistry	3/100	3%	2415	37.4 / 86	41.0/100 41.0%	▼ 38.0	4 / 13 / 8
JEE Advanced 89 / 171 · 52.0%							
Maths	34/54	63%	109	24.2 / 41	23.2/60 38.7%	▲ 24.2	10 / 4 / 1
Physics	21/60	35%	303	9.4 / 40	25.3/60 42.2%	▼ 7.2	7 / 8 / 2
Chemistry	34/57	60%	435	23.9 / 50	20.5/60 34.2%	▲ 25.5	11 / 5 / 0

Bar = your score % | campus average | best on campus Change = this test % − weekly-test average %

What this report says

1. **Maths** is your strongest JEE Mains subject: 53/96 (55%), All-India subject rank 103.
2. **Chemistry** needs the most work in JEE Mains: 3/100 (3%), subject rank 2415; you lost 21 of its 25 questions.
3. **Maths** is your strongest JEE Advanced subject: 34/54 (63%), All-India subject rank 109.
4. **Physics** needs the most work in JEE Advanced: 21/60 (35%), subject rank 303; you lost 10 of its 17 questions.
5. Biggest fall against your weekly-test average: **JEE Mains Physics** – 10% here against 49% on average.
6. Biggest improvement: **JEE Advanced Chemistry** – 60% here against 34% on average.
7. **20 questions** you lost were answered correctly by 70%+ of the batch's top 100 – about **94 marks** within easy reach (Mains Q28, Q31, Q32, Q51, Q52, Q56, Q60, Q61, Q64, Q66, Q68, Q70, Q71, Q74 · Advanced Q6, Q24, Q25, Q40, Q44, Q47).
8. Negative marking cost you **34 in JEE Mains and 8 in JEE Advanced**.
9. You left **15 in JEE Mains and 3 in JEE Advanced** questions blank, and answered 34 in JEE Mains and 17 in JEE Advanced wrong – both are marks to recover.

JEE Mains – question by question

66 / 296 · 22.3%
08 Oct 2026

2 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

Maths	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	~~20~~	21	22	23	24	25
Physics	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
Chemistry	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75

3 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 63 students of this program on campus.

Easy miss 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40-69% · **Tough one** below 40%

Maths

9 of 24 questions lost · **43** marks · 5 within reach · 4 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q13	Within reach	**Quadratic Equations**	Single correct · +4 / −1	Wrong	5	63%	27%
Q12	Within reach	**Sequences & Series**	Single correct · +4 / −1	Wrong	5	59%	19%
Q11	Within reach	**Quadratic Equations**	Single correct · +4 / −1	Wrong	5	46%	21%
Q25	Within reach	**Surds & Indices**	Numerical · +4 / −1	Wrong	5	44%	21%
Q4	Within reach	**Sequences & Series** – Identify AP / GP / HP from a surd expression	Single correct · +4 / −1	Wrong	5	42%	19%
Q22	Tough one	**Surds & Indices**	Numerical · +4 / −1	Skipped	4	25%	11%
Q23	Tough one	**Surds & Indices** – Cube root of a surd expression	Numerical · +4 / −1	Skipped	4	24%	13%
Q5	Tough one	**Quadratic Equations** – Range of a rational quadratic expression	Single correct · +4 / −1	Wrong	5	21%	6%
Q16	Tough one	**Quadratic Equations** – Roots in GP, condition on discriminant	Single correct · +4 / −1	Wrong	5	19%	3%

Physics

19 of 25 questions lost · **90** marks · 3 easy miss · 7 within reach · 9 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q31	Easy miss	**Vectors**	Single correct · +4 / −1	Wrong	5	80%	38%
Q28	Easy miss	**Vectors** – Resultant ⊥ smaller vector; find magnitudes	Single correct · +4 / −1	Skipped	4	77%	30%
Q32	Easy miss	**Kinematics**	Single correct · +4 / −1	Wrong	5	75%	29%
Q29	Within reach	**Vectors** – Angles between three vectors summing to zero	Single correct · +4 / −1	Wrong	5	62%	24%
Q40	Within reach	**Laws of Motion** – Velocity over fixed distance vs mass ($v \propto 1/\sqrt{m}$)	Single correct · +4 / −1	Wrong	5	62%	30%
Q44	Within reach	**Laws of Motion** – Normal reactions on a cylinder resting on edges	Single correct · +4 / −1	Wrong	5	53%	29%
Q50	Within reach	**Laws of Motion**	Numerical · +4 / −1	Skipped	4	44%	17%
Q27	Within reach	**Vectors** – Vector components under axis rotation	Single correct · +4 / −1	Wrong	5	43%	6%
Q30	Within reach	**Laws of Motion** – Tension / normal reaction in two-block systems (matrix match)	Single correct · +4 / −1	Wrong	5	42%	30%
Q35	Within reach	**Kinematics** – $a = kv^2$ – integrate to get v(s)	Single correct · +4 / −1	Skipped	4	41%	5%
Q41	Tough one	**Laws of Motion** – Newton's 3rd law – fan blowing on own sail	Single correct · +4 / −1	Wrong	5	31%	11%
Q42	Tough one	**Laws of Motion** – Extra tension when bird leaves the wire	Single correct · +4 / −1	Wrong	5	31%	14%
Q38	Tough one	**Kinematics** – Resistance $\propto v^2$ – time inside plank	Single correct · +4 / −1	Wrong	5	27%	11%
Q45	Tough one	**Laws of Motion** – Contact force between pushed blocks (statement type)	Single correct · +4 / −1	Wrong	5	27%	10%
Q36	Tough one	**Kinematics** – a(t) with modulus – area under a-t	Single correct · +4 / −1	Wrong	5	24%	5%
Q43	Tough one	**Laws of Motion** – Normal reaction across an inclined contact	Single correct · +4 / −1	Wrong	5	20%	10%
Q47	Tough one	**Vectors** – Net displacement from multi-leg journey	Numerical · +4 / −1	Skipped	4	15%	3%
Q48	Tough one	**Kinematics** – Distance from a-t graph	Numerical · +4 / −1	Wrong	5	15%	3%
Q49	Tough one	**Kinematics** – Free fall then deceleration – find height	Numerical · +4 / −1	Skipped	4	15%	2%

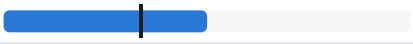
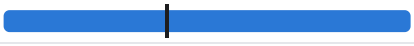
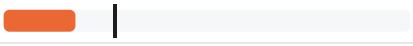
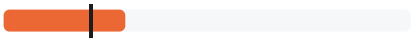
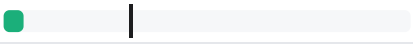
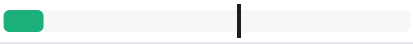
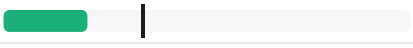
Chemistry

21 of 25 questions lost · **97** marks · 11 easy miss · 9 within reach · 1 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q64	Easy miss	**Periodic Classification**	Single correct · +4 / −1	Wrong	5	98%	76%
Q68	Easy miss	**Chemical Bonding**	Single correct · +4 / −1	Wrong	5	98%	83%
Q66	Easy miss	**Periodic Classification**	Single correct · +4 / −1	Skipped	4	91%	65%
Q71	Easy miss	**Mole Concept & Basics**	Numerical · +4 / −1	Skipped	4	90%	59%
Q51	Easy miss	**Mole Concept & Basics**	Single correct · +4 / −1	Wrong	5	85%	56%
Q61	Easy miss	**Atomic Structure**	Single correct · +4 / −1	Wrong	5	84%	40%
Q52	Easy miss	**Mole Concept & Basics**	Single correct · +4 / −1	Wrong	5	82%	62%
Q56	Easy miss	**Mole Concept & Basics**	Single correct · +4 / −1	Skipped	4	79%	56%
Q60	Easy miss	**Atomic Structure**	Single correct · +4 / −1	Skipped	4	79%	38%
Q70	Easy miss	**Chemical Bonding**	Single correct · +4 / −1	Wrong	5	72%	52%
Q74	Easy miss	**Periodic Classification**	Numerical · +4 / −1	Skipped	4	70%	46%
Q72	Within reach	**Mole Concept & Basics**	Numerical · +4 / −1	Skipped	4	69%	37%
Q54	Within reach	**Mole Concept & Basics** – Minimum molecular mass of insulin from 3.4% S	Single correct · +4 / −1	Wrong	5	67%	25%
Q55	Within reach	**Mole Concept & Basics** – Laws of chemical combination	Single correct · +4 / −1	Wrong	5	66%	17%
Q57	Within reach	**Atomic Structure** – Which electron jump gives the red line (smallest energy gap)	Single correct · +4 / −1	Wrong	5	66%	25%
Q53	Within reach	**Mole Concept & Basics** – Valence electrons in N^{3-} (mole → electrons)	Single correct · +4 / −1	Wrong	5	65%	25%
Q65	Within reach	**Periodic Classification** – Largest radius among isoelectronic ions	Single correct · +4 / −1	Wrong	5	64%	33%
Q73	Within reach	**Mole Concept & Basics**	Numerical · +4 / −1	Skipped	4	60%	22%
Q75	Within reach	**Chemical Bonding** – Hybrid orbitals + lone pairs in NH_3	Numerical · +4 / −1	Skipped	4	60%	30%
Q58	Within reach	**Atomic Structure**	Single correct · +4 / −1	Wrong	5	57%	16%
Q67	Tough one	**Chemical Bonding** – Strongest ionic bond (charge and size)	Single correct · +4 / −1	Wrong	5	32%	13%

4 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject.

CHAPTER	QS	YOUR MARKS	YOU VS CAMPUS AVERAGE	CAMPUS AVG	QUESTIONS YOU LOST
Maths					
Quadratic Equations	5	0/20		11%	Q5, Q11, Q13, Q16
Sequences & Series	5	10/20		33%	Q4, Q12
Surds & Indices	11	31/44		53%	Q22, Q23, Q25
Logarithms & Exponents	3	12/12		40%	– none
Physics					
Laws of Motion	8	-7/32		7%	Q30, Q40, Q41, Q42, Q43, Q44, Q45, Q50
Vectors	7	5/28		27%	Q27, Q28, Q29, Q31, Q47
Kinematics	10	12/40		21%	Q32, Q35, Q36, Q38, Q48, Q49
Chemistry					
Mole Concept & Basics	9	-5/36		32%	Q51, Q52, Q53, Q54, Q55, Q56, Q71, Q72, Q73
Atomic Structure	5	1/20		31%	Q57, Q58, Q60, Q61
Periodic Classification	5	2/20		58%	Q64, Q65, Q66, Q74
Chemical Bonding	6	5/24		34%	Q67, Q68, Q70, Q75

Bar = your % of the chapter's marks | campus average

5 JEE Mains · Maths – questions to revisit

9 questions · easy misses first
cropped from the question paper

Q13 Quadratic Equations

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **63%** got it right · Campus: **27%** (17 of 63)

Within reach
You: Wrong

13. The value of α for which the sum of the squares of the roots of equation $x^2-(\alpha-2)x-\alpha-1=0$ assume the least value, is

(A)1 (B)0 (C)3 (D)2

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q12 Sequences & Series

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **59%** got it right · Campus: **19%** (12 of 63)

Within reach
You: Wrong

12. If $4a^2+9b^2+16c^2=2(3ab+6bc+4ca)$, where a, b, c are nonzero numbers, then a, b, c are in

(A)A.P. (B)G.P. (C)H.P. (D)Neither A.P nor G.P nor H.P

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q11 Quadratic Equations

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **46%** got it right · Campus: **21%** (13 of 63)

Within reach
You: Wrong

11. The range of values of a for which all the roots of the equation $(a-1)\left(1+x+x^2\right)^2=(a+1)\left(1+x^2+x^4\right)$ are imaginary is

(A) $(-\infty,-2]$ (B) $(2,\infty)$ (C) $-2<a<2$ (D) $(-\infty,2]$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q25

Surds & Indices
JEE Mains · Maths · Numerical · marking +4 / −1 · marks lost 5
Top 100: **44%** got it right · Campus: **21%** (13 of 63)

Within reach
You: Wrong

25. If $\left(2+\sqrt{3}\right)^{x^2-7x+1}+\left(2-\sqrt{3}\right)^{x^2-7x-1}=\frac{2}{2-\sqrt{3}}$, then the sum of values of 'x' satisfying this equation is equal to_____ ?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q4

Sequences & Series – Identify AP / GP / HP from a surd expression
JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5
Top 100: **42%** got it right · Campus: **19%** (12 of 63)

Within reach
You: Wrong

4. If a, b, c are in A.P., then $\frac{1}{\sqrt{a}+\sqrt{b}}, \frac{1}{\sqrt{a}+\sqrt{c}}, \frac{1}{\sqrt{b}+\sqrt{c}}$ are in

(A)A.P. (B)G.P. (C)H.P. (D)Neither A.P nor G.P nor H.P

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q22

Surds & Indices
JEE Mains · Maths · Numerical · marking +4 / −1 · marks lost 4
Top 100: **25%** got it right · Campus: **11%** (7 of 63)

Tough one
You: Skipped

22. If $y=\frac{(p+1)^{\frac{1}{3}}+(p-1)^{\frac{1}{3}}}{(p+1)^{\frac{1}{3}}-(p-1)^{\frac{1}{3}}}$ then the value of the expression $y^3-3py^2+3y-p+3$ is_____?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q23 — Surds & Indices – Cube root of a surd expression

JEE Mains · Maths · Numerical · marking +4 / −1 · marks lost 4

Top 100: **24%** got it right · Campus: **13%** (8 of 63)

Tough one

You: Skipped

23. If $\sqrt[3]{38\sqrt{14}-100\sqrt{2}}=\sqrt{x}-\sqrt{y}$, then the value of $x-y$ is__?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q5 — Quadratic Equations – Range of a rational quadratic expression

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **21%** got it right · Campus: **6%** (4 of 63)

Tough one

You: Wrong

5. If x is real, then $\dfrac{x^2+2x+c}{x^2+4x+3c}$ can take all real values if

(A) $0 < c < 2$ (B) $0 \le c \le 1$

(C) $-1 < c < 1$ (D) $0 < c < 1$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q16 — Quadratic Equations – Roots in GP, condition on discriminant

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **19%** got it right · Campus: **3%** (2 of 63)

Tough one

You: Wrong

16. Let α, β be the roots of the equation $ax^2+bx+c=0$ and $\Delta=b^2-4ac$. If $\alpha+\beta, \alpha^2+\beta^2, \alpha^3+\beta^3$ are in G.P., then

(A) $\Delta=0$ (B) $\Delta\neq 0$ (C) $b\Delta=0$ (D) $c\Delta=0$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Mains · Physics – questions to revisit

19 questions · easy misses first
cropped from the question paper

Q31

Vectors
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5
Top 100: **80%** got it right · Campus: **38%** (24 of 63)

Easy miss
You: Wrong

31. A particle has position vector $\left(3\hat{i}-\hat{j}+2\hat{k}\right)$ metre at time t =0 . It moves with constant velocity $\left(-\hat{i}+\hat{j}-3\hat{k}\right)$ ms^{-1}. The position vector (in m) of the particle after 3 second is

(A) $2\hat{j}-7\hat{k}$ (B) $2\hat{i}-\hat{k}$ (C) $\hat{j}$ (D) $3\hat{k}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q28

Vectors – Resultant ⊥ smaller vector; find magnitudes
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4
Top 100: **77%** got it right · Campus: **30%** (19 of 63)

Easy miss
You: Skipped

28. The simple sum of two co-initial vectors is 16 units . Their vector sum is 8 units. The resultant of the vectors is perpendicular to the smaller vector . The magnitudes of the two vectors are

(A) 22units and 14 units (B) 4 units and 12 units
(C) 6 units and 10 units (D) 8 units and 12 units

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q32

Kinematics

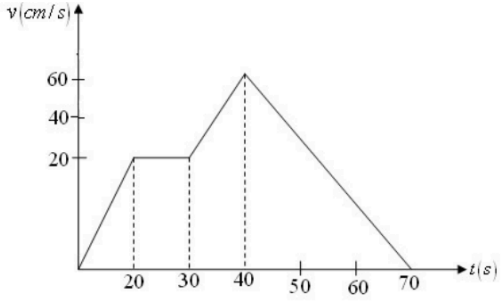
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **75%** got it right · Campus: **29%** (18 of 63)

Easy miss

You: Wrong

32. The velocity versus time curve of a moving points is given below. The maximum acceleration is



(A) $1\ \text{cm/s}^2$ (B) $2\ \text{cm/s}^2$ (C) $3\ \text{cm/s}^2$ (D) $4\ \text{cm/s}^2$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q29

Vectors – Angles between three vectors summing to zero

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **62%** got it right · Campus: **24%** (15 of 63)

Within reach

You: Wrong

29. Given: $\vec{a}+\vec{b}+\vec{c}=0$. Out of the three vectors $\vec{a},\vec{b}$ and $\vec{c}$, two are equal in magnitude. The magnitude of the third vector is $\sqrt{2}$ times that of either of the two having equal magnitude. The angles between the vector are

(A) $90^0,135^0,135^0$ (B) $30^0,60^0,90^0$

(C) $45^0,45^0,90^0$ (D) $45^0,60^0,90^0$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q40

Laws of Motion – Velocity over fixed distance vs mass (v ∝ 1/√m)

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **62%** got it right · Campus: **30%** (19 of 63)

Within reach

You: Wrong

40. A constant force acts on a mass m at rest. Velocity acquired in travelling a fixed distance is directly proportional to

(A) $\sqrt{m}$ (B) m (C) $1/\sqrt{m}$ (D) $1/m$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q44

Laws of Motion – Normal reactions on a cylinder resting on edges

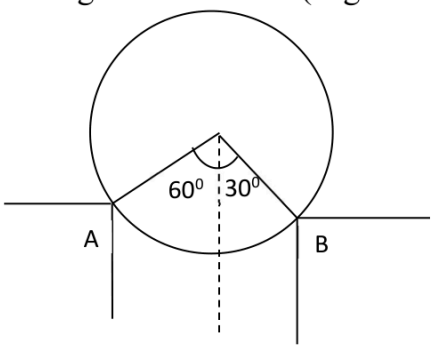
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **53%** got it right · Campus: **29%** (18 of 63)

Within reach

You: Wrong

44. A cylinder of mass M and radius R is resting on two corner edges A and B as shown in the figure. The normal reaction at the edges A and B are (neglect friction)



(A) $N_A = \sqrt{2}N_B$ (B) $N_B = \sqrt{3}N_A$ (C) $N_A = \sqrt{3}N_B$ (D) $N_B = \frac{2\sqrt{3}Mg}{5}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q50

Laws of Motion

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 4

Top 100: **44%** got it right · Campus: **17%** (11 of 63)

Within reach

You: Skipped

50. A constant force acts on a body of mass 1kg for 5 seconds, so that the body begins to move with zero initial velocity on a smooth horizontal plane. After 5 seconds the force ceases to act. In the next 5 second, the body describes 100m. What is force (in Newton) that acted on the body

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q27

Vectors – Vector components under axis rotation

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **43%** got it right · Campus: **6%** (4 of 63)

Within reach

You: Wrong

27. The components of a vector along x and y directions are (n+1) and 1 respectively . If the co-ordinate system is rotated by an angle $\theta = 60^0$, then the components change to n and 3. The value of n is

(A) 2 (B) $\cos 60^0$ (C) $\sin 60^0$ (D) 3.5

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q30

Laws of Motion – Tension / normal reaction in two-block systems (matrix match)

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **42%** got it right · Campus: **30%** (19 of 63)

Within reach

You: Wrong

30. Column-I gives four different situations involving two blocks of mass m_1 and m_2 placed in different ways on a smooth horizontal surface as shown. In each of the situations horizontal forces F_1 and F_2 are applied on blocks of mass m_1 and m_2 respectively and also $m_2 F_1 < m_1 F_2$. Match the statements in column I with corresponding results in column-II

Column-I	Column-II
(A)  . Both the blocks are connected by massless inelastic string. The magnitude of tension in the string is	(p) $\frac{m_1 m_2}{m_1+m_2}\left(\frac{F_1}{m_1}-\frac{F_2}{m_2}\right)$
(B)  . Both the blocks are connected by massless inelastic string. The magnitude of tension in the string is	(q) $\frac{m_1 m_2}{m_1+m_2}\left(\frac{F_1}{m_1}+\frac{F_2}{m_2}\right)$
(C)  . The magnitude of normal reaction between the blocks is	(r) $\frac{m_1 m_2}{m_1+m_2}\left(\frac{F_2}{m_2}-\frac{F_1}{m_1}\right)$
(D)  . The magnitude of normal reaction between the blocks is	(s) $m_1 m_2\left(\frac{F_1+F_2}{m_1+m_2}\right)$

(A) (A) – r ; (B) – q ; (C) – q ; (D) – r

(B) (A) – q ; (B) – r ; (C) – r ; (D) – q

(C) (A) – q ; (B) – r ; (C) – q ; (D) – r

(D) (A) – q ; (B) – p ; (C) – r ; (D) – r

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q35

Kinematics – a = kv² – integrate to get v(s)

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **41%** got it right · Campus: **5%** (3 of 63)

Within reach

You: Skipped

35. If a particle accelerates with $a = kv^2$ and initial velocity = u, then find velocity after displacement s

(A) $v = ue^{ks}$ (B) $v = u^{ks}e^{s}$ (C) $v = u^{k}e^{s}$ (D) $v = eu^{ks}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q41

Laws of Motion – Newton's 3rd law – fan blowing on own sail

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **31%** got it right · Campus: **11%** (7 of 63)

Tough one

You: Wrong

41. On a stationary sail boat, air is blown at the sails from a fan attached to the boat. The boat will:

(A) Move forward (B) move backward

(C) spin around (D) remain stationary

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q42

Laws of Motion – Extra tension when bird leaves the wire

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **31%** got it right · Campus: **14%** (9 of 63)

Tough one

You: Wrong

42. A bird dismounts on a stretched telegraph wire. The additional tension produced in the wire is

(A) Zero (B) greater than the weight of the bird

(C) Less than the weight of the bird (D) equal to the weight of the bird

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q38

Kinematics – Resistance ∝ v² – time inside plank

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **27%** got it right · Campus: **11%** (7 of 63)

Tough one

You: Wrong

38. Having gone through a plank of thickness h, a bullet changed its velocity from v_0 to v. Find the time of motion of the bullet in the plank assuming resistance force to be proportional to the square of the velocity

(A) $\left[\frac{h(v_0-v)}{vv_0 l\,n\frac{v_0}{v}}\right]$ (B) $\left[\frac{(v_0-v)}{hvv_0 l\,n\left(\frac{v_0}{v}\right)}\right]$ (C) $\left[\frac{vv_0 \ln\left(\frac{v_0}{v}\right)}{h(v_0-v)}\right]$ (D) $\left[\frac{\ln\frac{v_0}{v}}{vv_0(v_0-v)}\right]$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q45

Laws of Motion – Contact force between pushed blocks (statement type)

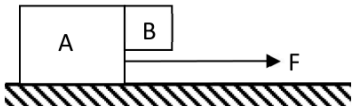
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **27%** got it right · Campus: **10%** (6 of 63)

Tough one

You: Wrong

45. Statement-1: Block A is moving on a horizontal surface towards right under action of force F. all surfaces are smooth. At the instant shown the force exerted by block A on block B is equal to net force on block B.



Statement-2: From Newton's third law, the force exerted by block A on B is equal in magnitude to force exerted by block B on A.

(A) Statement 1 is true, statement 2 is true, statement 2 is correct explanation for statement 1

(B) Statement 1 is true, statement 2 is true, statement 2 is not correct explanation for statement 1

(C) Statement 1 is true, statement 2 is false

(D) Statement 1 is false, statement 2 is true

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q36

Kinematics – a(t) with modulus – area under a-t

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **24%** got it right · Campus: **5%** (3 of 63)

Tough one

You: Wrong

36. Starting from rest a particle moves in a straight line with acceleration $a = \{2 + |t-2|\}\, m/s^2$. Velocity of the particle at the end of 4s will be

(A) 16 m/s (B) 20 m/s (C) 8 m/s (D) 12 m/s

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q43

Laws of Motion – Normal reaction across an inclined contact

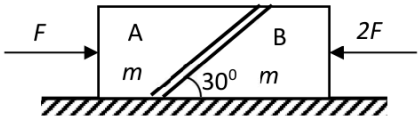
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **20%** got it right · Campus: **10%** (6 of 63)

Tough one

You: Wrong

43. Two blocks 'A" and 'B' each of mass 'm' are placed on a smooth horizontal surface. Two horizontal forces F and 2F are applied on both the blocks 'A' and 'B' respectively as shown in figure. The block A does not slide on block B. Then the normal reaction acting between the two blocks is



(A) F (B) $F/2$ (C) $F/\sqrt{3}$ (D) 3F

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q47

Vectors – Net displacement from multi-leg journey

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 4

Top 100: **15%** got it right · Campus: **3%** (2 of 63)

Tough one

You: Skipped

47. In going from one city to another , a car travels 75 km north, 60 km north –west and 20 km east. The magnitude(in km) of the vector drawn between the two cities is $\left(Take 1/\sqrt{2} = 0.7\right)$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q48

Kinematics – Distance from a-t graph

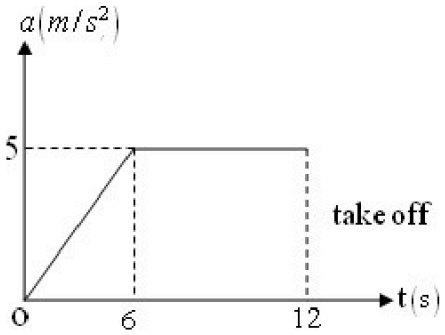
JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 5

Top 100: **15%** got it right · Campus: **3%** (2 of 63)

Tough one

You: Wrong

48. An experiment on the take off performance of an aeroplane shows that the acceleration varies as shown in the fig and it takes 12 sec to take off from a rest position.The distance along the runway covered by the aeroplane is(in m).



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q49

Kinematics – Free fall then deceleration – find height

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 4

Top 100: **15%** got it right · Campus: **2%** (1 of 63)

Tough one

You: Skipped

49. A parachutist after bailing out falls 50m without friction. When parachute opens, it decelerates at $2m/\sec^2$. He reaches the ground with a speed of 3m/s. The height he bail out is nearly (in m) (Take g=10m/sec^2) .

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Mains · Chemistry – questions to revisit

21 questions · easy misses first
cropped from the question paper

Q64 — Periodic Classification

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **98%** got it right · Campus: **76%** (48 of 63)

Easy miss
You: Wrong

64. Calcium is a member of -
(A) alkali metal (B) alkaline earth metal (C) p-block (D) d-block

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q68 — Chemical Bonding

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **98%** got it right · Campus: **83%** (52 of 63)

Easy miss
You: Wrong

68. Which of the following is an example of super octet molecule ?
(A) ClF_3 (B) PCl_5
(C) IF_7 (D) All the three

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q66 Periodic Classification

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 4
Top 100: **91%** got it right · Campus: **65%** (41 of 63)

Easy miss
You: Skipped

66.

Column-A		Column-B	
(A)	Vertical column in periodic table	(i)	IE_1
(B)	$A(g) + \text{Energy} \rightarrow A^+(g) + e^-$	(ii)	Period
(C)	Horizontal row in periodic table	(iii)	EA_2
(D)	$A(g) + e^- \rightarrow A^-(g) + \text{Energy}$	(iv)	Group
(E)	$A^+(g) + \text{Energy} \rightarrow A^{2+}(g) + e^-$	(v)	EA_1
(F)	$A^-(g) + e^- \rightarrow A^{2-}(g) + \text{Energy}$	(vi)	IE_2

A) [(A) → (i), (B) → (iv) , (C) → (iii) , (D) → (v) ,(E) → (vi) , (F) → (ii)]
B) [(A) → (iv), (B) → (iii) , (C) → (ii) , (D) → (vi) ,(E) → (v) , (F) → (i)]
C) [(A) → (iv), (B) → (i) , (C) → (ii) , (D) → (v) ,(E) → (vi) , (F) → (iii)]
D) [(A) → (v), (B) → (i) , (C) → (ii) , (D) → (iv) ,(E) → (iii) , (F) → (vi)]

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q71 Mole Concept & Basics

JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 4
Top 100: **90%** got it right · Campus: **59%** (37 of 63)

Easy miss
You: Skipped

71. The number of moles of oxygen atoms present in 0.50 mol of $Al_2(SO_4)_3$ is:

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q51 Mole Concept & Basics

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **85%** got it right · Campus: **56%** (35 of 63)

Easy miss
You: Wrong

51. In a hydrocarbon, there is 3gm of carbon per gm of hydrogen present in the molecule. Therefore, molecular formula of the hydrocarbon is-

(A) CH_4 (B) C_2H_6
(C) C_3H_8 (D) C_4H_{10}

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q61

Atomic Structure
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **84%** got it right · Campus: **40%** (25 of 63)

Easy miss
You: Wrong

61. **Assertion:** Emitted radiation will fall in visible range when an electron jump from n = 4 to n = 2 in H atom.
Reason: Balmer series radiation belongs to visible range for hydrogen atom.
(A)If both (A) and (R) are true, and (R) is the correct explanation of (A).
(B)If both (A) and (R) are true but (R) is not the correct explanation of (A).
(C)If (A) is true but (R) is false.
(D) If (A) is false but (R) is true.

Why did I lose this question? (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q52

Mole Concept & Basics
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **82%** got it right · Campus: **62%** (39 of 63)

Easy miss
You: Wrong

52. The mass of 2gram atoms of calcium (Relative atomic mass = 40)
(A) 2 g (B) 0.05 g (C) 0.5 g (D) 80 g

Why did I lose this question? (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q56

Mole Concept & Basics
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 4
Top 100: **79%** got it right · Campus: **56%** (35 of 63)

Easy miss
You: Skipped

56. **Assertion :** 1 mole of sulphuric acid contains 32 g each of sulphur and oxygen element.
Reason : 1 mole of sulphuric acid weights 98 g
(A)If both (A) and (R) are true, and (R) is the correct explanation of (A).
(B)If both (A) and (R) are true but (R) is not the correct explanation of (A).
(C)If (A) is true but (R) is false.
(D) If (A) is false but (R) is true.

Why did I lose this question? (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q60 **Atomic Structure**
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 4
Top 100: **79%** got it right · Campus: **38%** (24 of 63)

Easy miss
You: Skipped

60. When a gold sheet is bombarded by a beam of α-particles only a few of them deflected, whereas most go straight undeflected, because of
(A) The force attraction exerted on α-particles by the oppositely charged electrons is not sufficient.
(B) A nucleus has much smaller volume than that of an atom.
(C) The force of repulsion acting on the fast moving α-particles is very small.
(D) The neutrons in the nucleus do not have any effect on α-particles

Why did I lose this question? (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q70 **Chemical Bonding**
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **72%** got it right · Campus: **52%** (33 of 63)

Easy miss
You: Wrong

70. **Assertion :** Solid NaCl is conducting in nature
Reason : NaCl cantain Na^+, Cl^-.
(A)If both (A) and (R) are true, and (R) is the correct explanation of (A).
(B)If both (A) and (R) are true but (R) is not the correct explanation of (A).
(C)If (A) is true but (R) is false.
(D) If (A) is false but (R) is true.

Why did I lose this question? (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q74 **Periodic Classification**
JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 4
Top 100: **70%** got it right · Campus: **46%** (29 of 63)

Easy miss
You: Skipped

74. An element X belongs to the 3rd period and has 5 valence electrons. Another element Y belongs to the same period and has 2 valence electrons.
Find the sum of the atomic numbers of X and Y.

Why did I lose this question? (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q72

Mole Concept & Basics
JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 4
Top 100: **69%** got it right · Campus: **37%** (23 of 63)

Within reach
You: Skipped

72. At STP, 5 Moles of N_2 reacts completely with excess H_2:
$N_2 + 3H_2 \rightarrow 2NH_3$.
The mass of NH_3 formed is _____ g.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q54

Mole Concept & Basics – Minimum molecular mass of insulin from 3.4% S
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **67%** got it right · Campus: **25%** (16 of 63)

Within reach
You: Wrong

54. Insulin contains 3.4% sulphur. What will be the minimum molecular weight of insulin ?
(A) 94.176 (B) 1884 (C) 941.176 (D) 976

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q55

Mole Concept & Basics – Laws of chemical combination
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **66%** got it right · Campus: **17%** (11 of 63)

Within reach
You: Wrong

55. 12 g carbon combines with 64 g sulphur to form CS_2. 12 g carbon also combines with 32 g oxygen to form CO_2. 10 g sulphur combines with 10 g oxygen to form SO_2. These data illustrate the -
(A) Law of multiple proportions
(B) Law of definite proportions
(C) Law of reciprocal proportions
(D) Law of gaseous volumes.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q57 — Atomic Structure – Which electron jump gives the red line (smallest energy gap)

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **66%** got it right · Campus: **25%** (16 of 63)

Within reach

You: Wrong

57. Suppose that a hypothetical H-like atom gives a red Green, blue & violet line spectrum. Which jump according to figure would give off the red spectral line.



(A) 3 – 1 (B) 2 → 1 (C) 4 – 1 (D) 3 →2

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q53 — Mole Concept & Basics – Valence electrons in N^{3-} (mole → electrons)

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **65%** got it right · Campus: **25%** (16 of 63)

Within reach

You: Wrong

53. If N_A is Avogadro number, then the number of valence electrons in 4.2 g of N^{3-} ions is

(A) 2.4 N_A (B) 4.2 N_A (C) 1.6 N_A (D) 3.2 N_A

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q65 — Periodic Classification – Largest radius among isoelectronic ions

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **64%** got it right · Campus: **33%** (21 of 63)

Within reach

You: Wrong

65. Which of the following has largest radius ?

(A) Mg^{2+} (B) Na^{+} (C) O^{2-} (D) F^{-}

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q73 — Mole Concept & Basics

JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 4

Top 100: **60%** got it right · Campus: **22%** (14 of 63)

Within reach

You: Skipped

73. 3.011×10^{23} molecules of CO_2 contain $x \times 10^{23}$ atoms in total. Find x.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q75 — Chemical Bonding – Hybrid orbitals + lone pairs in NH_3

JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 4

Top 100: **60%** got it right · Campus: **30%** (19 of 63)

Within reach

You: Skipped

75. According to VBT, consider the molecule NH_3. The central nitrogen atom undergoes hybridisation before bond formation.
If the number of hybrid orbitals formed is x, and the number of lone pairs present on nitrogen is y, find:
$x + y$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q58 — Atomic Structure

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **57%** got it right · Campus: **16%** (10 of 63)

Within reach

You: Wrong

58. The ratio of the radii of first orbits of H, He^+ and Li^{2+} is -
(A) 1 : 2 : 3 (B) 6 : 3 : 2 (C) 1 : 4 : 9 (D) 9 : 4 : 1

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q67 **Chemical Bonding – Strongest ionic bond (charge and size)**

JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5

Top 100: **32%** got it right · Campus: **13%** (8 of 63)

Tough one

You: Wrong

67. Which combination will give strongest ionic bond?

(A) K^+ and Cl^- (B) K^+ and O^{2-}

(C) Ca^{2+} and Cl^- (D) Ca^{2+} and O^{2-}

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Advanced – question by question

89 / 171 · 52.0%
09 Oct 2026

6 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

Maths	1	2	3	4	5	6	7	8	9	10	11	12	13	~~14~~	15	16	~~17~~
Physics	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
Chemistry	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	~~50~~	51

7 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 63 students of this program on campus.

Easy miss 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40-69% · **Tough one** below 40%

Maths

5 of 15 questions lost · **20** marks · 1 easy miss · 2 within reach · 2 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q6	Easy miss	**Quadratic Equations** – Other root of $4x^2 + 2x - 1 = 0$ in terms of α	Multiple correct · +4 / −2	Wrong	6	82%	67%
Q10	Within reach	**Quadratic Equations** – Roots of two linked quadratics	Integer · +4 / 0	Wrong	4	50%	25%
Q15	Within reach	**Sequences & Series** – AP: ratio of sums of 14 terms	Paragraph · +3 / 0	Wrong	3	42%	8%
Q16	Tough one	**Logarithms** – Log equations – A + B + C	Paragraph · +3 / 0	Wrong	3	18%	3%
Q13	Tough one	**Quadratic Equations** – Spot the polynomial identity, then sum P(1) … P(2023)	Integer · +4 / 0	Skipped	4	– no data*	2%

Physics

10 of 17 questions lost · **38** marks (+1 on partly-correct answers) · 2 easy miss · 1 within reach · 7 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q25	Easy miss	**Vectors**	Integer · +4 / 0	Wrong	4	84%	37%
Q24	Easy miss	**Laws of Motion** – Tensions in two inclined strings	Multiple correct · +4 / −2	Wrong	6	70%	44%
Q30	Within reach	**Kinematics** – a from v(t) at a given position	Integer · +4 / 0	Wrong	4	43%	8%
Q28	Tough one	**Laws of Motion** – Spring-connected blocks – acceleration	Integer · +4 / 0	Wrong	4	25%	13%
Q33	Tough one	**Kinematics** – Train passage – retardation to avoid collision	Paragraph · +3 / 0	Skipped	3	15%	8%
Q34	Tough one	**Kinematics** – Train passage – time of retardation	Paragraph · +3 / 0	Skipped	3	15%	10%
Q27	Tough one	**Laws of Motion** – Normal reaction between pushed blocks	Integer · +4 / 0	Wrong	4	11%	2%
Q31	Tough one	**Vectors** – Carrom passage – displacement centre → edge	Paragraph · +3 / 0	Wrong	3	7%	2%
Q29	Tough one	**Kinematics**	Integer · +4 / 0	Wrong	4	– no data*	6%
Q32	Tough one	**Vectors** – Carrom passage – displacement edge → hole	Paragraph · +3 / 0	Wrong	3	6%	2%

Chemistry

5 of 16 questions lost · 23 marks · 3 easy miss · 1 within reach · 1 tough one

Q	PRIORITY	CHAPTER – WHAT IT TESTS	TYPE	YOU	MARKS LOST	TOP 100 RIGHT	CAMPUS RIGHT
Q40	Easy miss	Atomic Structure	Multiple correct · +4 / −2	Wrong	6	82%	44%
Q44	Easy miss	Atomic Structure	Integer · +4 / 0	Wrong	4	82%	51%
Q47	Easy miss	Mole Concept & Basics – Mass of water from 8 g H_2 + 32 g O_2 (limiting reagent)	Integer · +4 / 0	Wrong	4	72%	32%
Q39	Within reach	Mole Concept & Basics – Meaning of 1 gram-atom of nitrogen	Multiple correct · +4 / −2	Wrong	6	66%	22%
Q51	Tough one	Atomic Structure – Hydrogen-like ion – find Z from photon energies	Paragraph · +3 / 0	Wrong	3	0%	5%

8 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject. Chapter marks are approximate (≈): partly-correct answers cannot be traced to a question.

CHAPTER	QS	YOUR MARKS	YOU VS CAMPUS AVERAGE	CAMPUS AVG	QUESTIONS YOU LOST
Maths					
Quadratic Equations	7	≈ 12/26		43%	Q6, Q10, Q13
Logarithms	4	≈ 10/13		50%	Q16
Sequences & Series	4	≈ 12/15		56%	Q15
Physics					
Laws of Motion	4	≈ 2/16		6%	Q24, Q27, Q28
Vectors	5	≈ 7/17		23%	Q25, Q31, Q32
Kinematics	8	≈ 13/27		24%	Q29, Q30, Q33, Q34
Chemistry					
Atomic Structure	4	≈ 1/14		33%	Q40, Q44, Q51
Mole Concept & Basics	9	≈ 23/33		44%	Q39, Q47
Chemical Bonding	1	≈ 3/3		41%	– none
Periodic Classification	2	≈ 7/7		68%	– none

Bar = your % of the chapter's marks | campus average

9 JEE Advanced · Maths – questions to revisit

5 questions · easy misses first
cropped from the question paper

Q6 — Quadratic Equations – Other root of 4x² + 2x – 1 = 0 in terms of α

JEE Advanced · Maths · Multiple correct · marking +4 / −2 · marks lost 6
Top 100: **82%** got it right · Campus: **67%** (42 of 63)

Easy miss
You: Wrong

6. If α is a root of the equation $4x^2+2x-1=0$, then it's other root is given by

A) $4\alpha^3-3\alpha$ B) $4\alpha^3+3\alpha$ C) $\alpha-\frac{1}{2}$ D) $-\alpha-\frac{1}{2}$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q10 — Quadratic Equations – Roots of two linked quadratics

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4
Top 100: **50%** got it right · Campus: **25%** (16 of 63)

Within reach
You: Wrong

10. Let a and b be roots of the equation $x^2-10cx-11d=0$ and the roots of $x^2-10ax-11b=0$ are c and d; then the value of $\frac{a+b+c+d}{605}$ is____?(Given $a\neq b\neq c\neq d\neq 0$)

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q15 — Sequences & Series – AP: ratio of sums of 14 terms

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3
Top 100: **42%** got it right · Campus: **8%** (5 of 63)

Within reach
You: Wrong

Sum of n terms of two different Arithmetic progressions are in the ratio 5n+4:9n+6 then
15. The ratio of the sum of their first 14 terms is Q, then [100Q] is _____(where [.] is G.I.F)

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q16

Logarithms – Log equations – A + B + C

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **18%** got it right · Campus: **3%** (2 of 63)

Tough one

You: Wrong

PARAGRAPH-II

Let **A** denote the product xyz where x, y, z are solutions of the system of equations

$\log_3 x = \log_3 5 - \log_3 7, \log_5 y = \log_5 7 - \log_5 3, \log_7 z = \log_7 3 - \log_7 5$

Let **B** denote the sum of square of the solutions of the equation

$\log_2\left(\log_2 x^6 - 3\right) - \log_2\left(\log_2 x^4 - 5\right) = \log_2 3$

Let **C** denote the characteristic of the number K, where

$K = \log_2\left(\log_2 3\right) - \log_2\left(\log_4 3\right) + \log_2\left(\log_4 5\right) - \log_2\left(\log_6 5\right) + \log_2\left(\log_6 7\right) - \log_2\left(\log_8 7\right)$

16. The value of A+B+C is____?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q13

Quadratic Equations – Spot the polynomial identity, then sum P(1) ... P(2023)

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4

Top 100: **no data** got it right · Campus: **2%** (1 of 63)

Tough one

You: Skipped

13. Let $P(x) = \left(1+\sqrt{2}\right)\left(1+\sqrt{3}\right)\left(x-\sqrt{2}\right)\left(x-\sqrt{3}\right) + \left(1+\sqrt{3}\right)\left(\sqrt{3}+\sqrt{2}\right)(x-1)\left(x-\sqrt{2}\right)$
$-2\left(1+\sqrt{2}\right)\left(\sqrt{3}+\sqrt{2}\right)(x-1)\left(x-\sqrt{3}\right)\ \forall x \in R$. Then the value of
$P(1)+P(2)+P(3)+\ldots P(2023)$ is_____?

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Advanced · Physics – questions to revisit

10 questions · easy misses first
cropped from the question paper

Q25 Vectors

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4
Top 100: **84%** got it right · Campus: **37%** (23 of 63)

Easy miss
You: Wrong

25. The vectors $\vec{A}= 5\hat{i}- 7\hat{j}- 2\hat{k}$ and $\vec{B}= 2\hat{i}+ 2\hat{j}- c\hat{j}$ are perpendicular. Find the value of C.

Why did I lose this question? (write the reason and the correct approach)

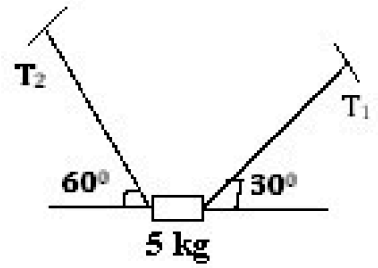
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q24 Laws of Motion – Tensions in two inclined strings

JEE Advanced · Physics · Multiple correct · marking +4 / −2 · marks lost 6
Top 100: **70%** got it right · Campus: **44%** (28 of 63)

Easy miss
You: Wrong

24. A body of mass 5 kg is suspended by the strings making angles 60^0 and 30^0 with the horizontal as shown in the figure($g = 10ms^{-2}$). Then



A) $T_1 = 25N$ B)$T_2 = 25N$ C) $T_1 = 25\sqrt{3}N$ D) $T_2 = 25\sqrt{3}N$

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q30 Kinematics – a from v(t) at a given position

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4
Top 100: **43%** got it right · Campus: **8%** (5 of 63)

Within reach
You: Wrong

30. A body starts from origin and moves along x axis such that at any instant velocity is $v_t=4t^3-2t$ where t is in second and v_t is in ms^{-1}. The acceleration of the particle when it is 2m from the origin is $(20+K)ms^{-2}$, where K is

Why did I lose this question? (write the reason and the correct approach)

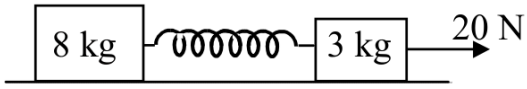
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q28

Laws of Motion – Spring-connected blocks – acceleration
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4
Top 100: **25%** got it right · Campus: **13%** (8 of 63)

Tough one
You: Wrong

28. If the acceleration of 3 kg block is 4 m/s^2 to the left. What is the magnitude of acceleration of 8 kg block.



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q33

Kinematics – Train passage – retardation to avoid collision
JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3
Top 100: **15%** got it right · Campus: **8%** (5 of 63)

Tough one
You: Skipped

PASSAGE-II:

The driver of a train A moving with a uniform speed of 144 km/h sights another train B, 1km ahead of him. The train B is moving with a uniform speed of 108 km/h. the driver of the train A immediately applies brakes producing a constant retardation and just manages to avoid a collision.

33. What is the retardation of the train A(in ms^{-2})_____

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q34

Kinematics – Train passage – time of retardation
JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3
Top 100: **15%** got it right · Campus: **10%** (6 of 63)

Tough one
You: Skipped

PASSAGE-II:

The driver of a train A moving with a uniform speed of 144 km/h sights another train B, 1km ahead of him. The train B is moving with a uniform speed of 108 km/h. the driver of the train A immediately applies brakes producing a constant retardation and just manages to avoid a collision.

34. For how long is this retardation produced(in sec)_____

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q27

Laws of Motion – Normal reaction between pushed blocks

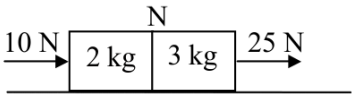
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **11%** got it right · Campus: **2%** (1 of 63)

Tough one

You: Wrong

27. The normal reaction between 2 kg and 3 kg block is



Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q31

Vectors – Carrom passage – displacement centre → edge

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **7%** got it right · Campus: **2%** (1 of 63)

Tough one

You: Wrong

PASSAGE-1:

A Corrom board $(4\,ft \times 4\,ft\ square)$ has queen at the centre. The queen , hit by the striker moves to the front edge, rebounds and goes in the hole behind the striking line.

Answer the following question.

31. The magnitude of the displacement of the queen from the centre to the front edge is $\frac{2}{3}\sqrt{a}\,ft$, the value of a is _____

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q29

Kinematics

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **no data** got it right · Campus: **6%** (4 of 63)

Tough one

You: Wrong

29. A lead ball is dropped freely into a lake from a diving board 5 m above the water. It strikes the water with a certain velocity and then reaches the bottom with the same constant velocity in 5 second after it is dropped.

Take g=10 ms^{-2} calculate the average velocity of the ball in ms^{-1} just before reaches to the bottom from the freely dropped point .

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q32 **Vectors – Carrom passage – displacement edge → hole**

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **6%** got it right · Campus: **2%** (1 of 63)

Tough one

You: Wrong

PASSAGE-1:

A Corrom board $(4\,ft \times 4\,ft\ square)$ has queen at the centre. The queen , hit by the striker moves to the front edge, rebounds and goes in the hole behind the striking line. Answer the following question.

32. The magnitude of the displacement of the queen from the front edge to the hole is $\frac{4}{3}\sqrt{b}\ ft$, the value of b is_____

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

JEE Advanced · Chemistry – questions to revisit

5 questions · easy misses first
cropped from the question paper

Q40 — Atomic Structure

JEE Advanced · Chemistry · Multiple correct · marking +4 / −2 · marks lost 6
Top 100: **82%** got it right · Campus: **44%** (28 of 63)

Easy miss
You: Wrong

40. Which of the following properties is/are proportional to the energy of the electromagnetic radiation ?

(A) Frequency
(B) Wave number
(C) Wavelength
(D) Number of photons

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q44 — Atomic Structure

JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4
Top 100: **82%** got it right · Campus: **51%** (32 of 63)

Easy miss
You: Wrong

44. An atom has 17 protons and 18 neutrons. It gains one electron. Find the sum of the mass number and the number of electrons in the resulting ion.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q47 — Mole Concept & Basics – Mass of water from 8 g H_2 + 32 g O_2 (limiting reagent)

JEE Advanced · Chemistry · Integer · marking +4 / 0 · marks lost 4
Top 100: **72%** got it right · Campus: **32%** (20 of 63)

Easy miss
You: Wrong

47. For $2H_2 + O_2 \rightarrow 2H_2O$, 8 g H_2 reacts with 32 g O_2. Find the mass of water formed.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q39 — Mole Concept & Basics – Meaning of 1 gram-atom of nitrogen

JEE Advanced · Chemistry · Multiple correct · marking +4 / −2 · marks lost 6

Top 100: **66%** got it right · Campus: **22%** (14 of 63)

Within reach

You: Wrong

39. 1 g atom of nitrogen represents:

(A) 6.02×10^{23} N_2 molecules

(B) 22.4 litre of N_2 at N.T.P.

(C) 11.2 litre of N_2 at N.T.P.

(D) 14 g of nitrogen

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

Q51 — Atomic Structure – Hydrogen-like ion – find Z from photon energies

JEE Advanced · Chemistry · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **0%** got it right · Campus: **5%** (3 of 63)

Tough one

You: Wrong

PARAGRAPH-II

When electron jumps from lower orbit to higher orbit energy is gained and when electron jumps from higher orbit to lower orbit energy is released.

51. A hydrogen like species (atomic number Z) is present in a higher excited state of quantum number n. This excited atom can make a transition to the first excited state by successive emission of two photons of energies 10.20 eV and 17.0 eV respectively. Alternatively the atom from the same excited state can make a transition to the second excited state by successive emission of two photons of energy 4.25 eV and 5.95 eV respectively. Determine the value of Z.

Why did I lose this question? (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

10 How to read this report

Where the numbers come from. Marks, percentages, All-India / State / Campus ranks, subject marks, subject All-India ranks and the wrong / skipped questions are the official results in the institute's *All India Marks Analysis* (graded with the final answer key). Maximum marks leave out deleted questions — JEE Mains: Q20 — JEE Advanced: Q14, Q17, Q50. **Program rank** = rank in Pragya G8 (EXI) Achievers across all campuses, from the batch's question-wise sheet († JEE Advanced: that sheet was graded before the final key, so this rank may differ slightly). **Top 100** = the batch's top 100 across all campuses, from the same question-wise sheet; * where it has no data for a question, the campus figure decides the priority. In JEE Advanced that sheet predates the final key, so its figures for the revised questions Q25, Q48, Q51 use the old answers. **Campus** = all students of this program on campus. **Priority:** Easy miss = 70%+ of the top 100 got it right; Within reach = 40–69%; Tough one = below 40%. **Marks lost** on a question = its marks plus the negative mark if answered wrong, its marks if skipped. In JEE Advanced, 1 mark went on multiple-correct answers that were only partly correct. **Weekly-test average** = your average over the weekly Mains tests (WTM) or Advanced tests (WTA) before this Part Test, from the campus averages sheet. The weekly Advanced tests are marked out of 240; their averages are scaled to 180 to match this paper (× 180/240, so 80 per subject becomes 60). Percentages are unchanged.